C21 Acids, Bases and Buffers – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Definitions and pH **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Brønsted–Lowry acid				Proton donor
2	Brønsted–Lowry base				Proton acceptor
3	Conjugate acid–base pair				Species differing by one H^+
4	Strong acid				Fully dissociates
5	Weak acid				Partially dissociates
6	pH equation				$pH = -\log[H^+]$
7	$[H^+]$ from pH				$[H^+] = 10^{-pH}$
8	Ionic product of water (K_w)				$K_w = [H^+][OH^-]$
9	Value of K_w at 298 K				1.00×10^{-14} mol^2 dm^{-6}
10	$[H^+]$ of a strong base solution				$K_w \div [OH^-]$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C21 Acids, Bases and Buffers – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 2: Weak Acids **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	K_a expression for HA				$[H^+][A^-] \div [HA]$
2	Units of K_a				mol dm^{-3}
3	pK_a				$-\log K_a$
4	K_a from pK_a				10^{-pKa}
5	$[H^+]$ of a weak acid				$\sqrt{(K_a[HA])}$
6	Assumption 1 for weak acid pH				$[H^+] = [A^-]$
7	Assumption 2 for weak acid pH				[HA] at equilibrium = initial [HA]
8	Larger K_a means				Stronger acid
9	Effect of raising temperature on K_w				Increases
10	Pure water at any temperature				$[H^+] = [OH^-]$, so neutral

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C21 Acids, Bases and Buffers – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 3: Buffers **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Buffer				Resists pH change when small amounts of acid or base are added
2	Acidic buffer				Weak acid + its salt
3	Basic buffer				Weak base + its salt
4	Example of a basic buffer				NH_3 and NH_4Cl
5	$[H^+]$ of an acidic buffer				$K_a \times [HA] \div [A^-]$
6	pH when $[HA] = [A^-]$				$pH = pK_a$
7	Buffer in blood				H_2CO_3 / HCO_3^-
8	Conjugate base of CH_3COOH				CH_3COO^-
9	Conjugate acid of NH_3				NH_4^+
10	Species that removes added H^+ in an acidic buffer				A^-

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C21 Acids, Bases and Buffers – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 4: pH Curves and Indicators **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Equivalence point				Moles of acid and base exactly react
2	pH at equivalence: strong acid–strong base				7
3	pH at equivalence: weak acid–strong base				Above 7
4	pH at equivalence: strong acid–weak base				Below 7
5	pH at half-equivalence (weak acid)				pK_a
6	Indicator choice rule				Changes colour within the vertical section
7	Indicator for weak acid–strong base				Phenolphthalein
8	Indicator for strong acid–weak base				Methyl orange
9	Indicator for weak acid–weak base				None suitable
10	Starting pH of 0.1 mol dm^{-3} HCl				1

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.